

Project Design Phase-II
Technology Stack (Architecture & Stack)

Date	31 March 2026
Team ID	
Project Name	Online Complaint Registration and Management System
Maximum Marks	4 Marks

Technical Architecture:

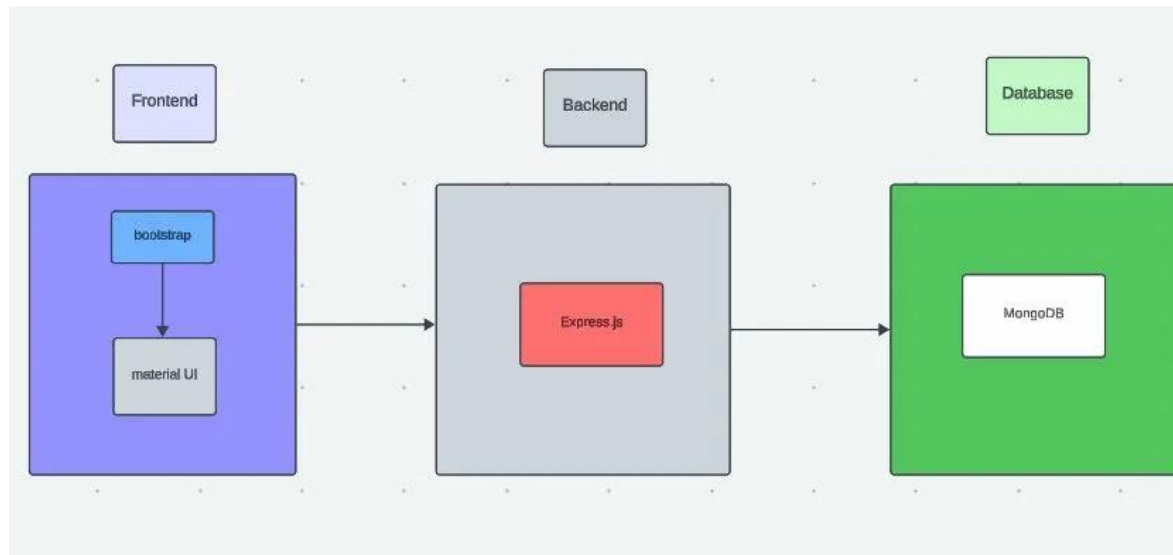


Table-1 : Components & Technologies:

S.No	Component	Description	Technology
1.	User Interface	Web-based UI for users, agents, and admin	React.js, Bootstrap, Material UI
2.	Application Logic-1	Authentication (login, register, JWT handling)	Node.js, Express.js, JWT
3.	Application Logic-2	Complaint handling (create, update, track)	Express.js, Mongoose
4.	Application Logic-3	Role-based access (User, Agent, Admin)	Express Middleware
5.	Database	Stores users, complaints, status	MongoDB (NoSQL)
6.	Cloud Database	Hosted database service	MongoDB Atlas
7.	File Storage	Stores complaint images	Local filesystem (uploads/)
8.	External API-1	HTTP communication between frontend & backend	REST APIs (Axios)
9.	Machine Learning Model	Not used in current system	N/A
10.	Infrastructure (Server / Cloud)	Backend server and frontend hosting	Localhost / Cloud (Node.js server)

Table-2: Application Characteristics:

S.No	Characteristics	Description	Technology
1.	Open-Source Frameworks	Frameworks used in development	React.js, Express.js, MongoDB
2.	Security Implementations	Authentication and data protection	JWT Authentication, bcrypt password hashing
3.	Scalable Architecture	Supports increasing users and complaints	MVC Architecture, REST APIs
4.	Availability	System accessible anytime	Node.js server, MongoDB Atlas
5.	Performance	Fast API response and data retrieval	Express.js, Indexed MongoDB queries